James William Hanzell

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Education

University of Toronto Toronto, ON

Honours BSc, Mathematics Major, Computer Science and Statistics Minors *Aug. 2024 – May 2028*

- **Coursework** Algorithm Design, Data Structures, Software Design, Systems Programming, Computer Organization, Linear Algebra II, Multivariable Calculus, Differential Equations, Statistics
- Executive at UofT Machine Intelligence Student Team, Computer Science Student Union and UofT AI Agents. Ran technical workshops and mentored first-years, reaching 1,000+ students.
- Available for a full-time internship **May to Aug. 2027**.

Experience

Lirvana Labs June 2026 – Present

Software Engineer Intern *Remote (Silicon Valley, CA)*

- Built **Concept Pages** end to end, AI-generated interactive lessons with manipulable 3D and 2D diagrams, worked solutions, and an adaptive quiz that visualizes each student's specific mistake and re-tests it with new numbers, helping close a **$200K** enterprise deal.
- Shipped two game types (Drag & Sort, Categorize) and Guided Notes, study material with auto-generated diagrams built from a teacher's curriculum, across all 8 activity types by extending the shared content model and AI-generation pipeline. **15+ reviewed PRs** in three months.
- Built a sortable class dashboard with student join codes and per-question difficulty and Depth-of-Knowledge metadata, and rebuilt the analytics UI in a production Next.js 16, React 19 and **Rust (gRPC, PostgreSQL) monorepo**.
- Replaced a three-panel curriculum picker with one breadcrumb browser on existing gRPC endpoints, with no new Protocol Buffers or schema changes, localized into 5 languages with a rewritten test suite.
- Fixed a drag-and-drop defect across 6+ game components and closed **30+ high-priority bugs** in generation, LaTeX rendering and game mechanics.

Ernst & Young (EY) June 2025 – Aug. 2025

AI Development Intern *Toronto, ON*

- Cut client-proposal drafting time by ∼**85%** (4 hrs → 5 min) with a Python ETL pipeline over 25,000+ WHO records feeding an LLM document-generation workflow, deployed in a client-facing pilot.
- Designed the generation pipeline as pluggable LLM revision stages so one codebase serves multiple proposal formats.
- Cut AI-interview response latency below **3s** by streaming speech-to-text and model inference concurrently instead of sequentially.

Projects

Vortex | *Next.js 16, TypeScript, Sanity, MCP, Vercel AI SDK, Groq, Clerk, PostHog* | Live Site | GitHub Sep. 2026 – Present

- Built an AI learning platform that turns a plain-English goal into a sequenced YouTube course. An LLM designs the syllabus and a search query per lesson, the YouTube Data API returns safe, embeddable candidates, and a second LLM pass rejects off-topic matches.
- Built hallucination-proof search over 120 lessons with the Sanity Context MCP. The model writes GROQ and returns only lesson IDs and timestamps, every displayed field is re-read from the database, and results deep-link to the exact second.
- Kept search fast under an 8K-tokens/min free-tier limit with a parallel keyword pass, a 12s model budget with keyword-only fallback, and capacity-aware fallback across 3 models.
- Secured all writes behind Clerk-authenticated server routes with Zod validation and compare-and-set database commits that hold plan limits under concurrent requests, and fixed a production stale-cache bug with an HMAC-signed revalidation webhook.

Coven | *React, TypeScript, Leaflet, SunCalc, Web Workers, Vercel* | Live Site | GitHub Oct. 2026

- **Winner, Hack-or-Treat Hackathon** (team of 4, 5 hours). Built the app shell and a sun-safe walking router for the UofT St. George campus.
- Projected shadows for ∼1,200 buildings from City of Toronto LiDAR 3D massing data and live sun position, then ran Dijkstra over a campus walking graph weighted by metres spent in direct sun.
- Kept clicks under 100ms by moving shadow computation into a Web Worker with 15-minute prefetching, and drawing all shadows as one canvas path instead of 1,200 Leaflet layers.

Technical Skills

Languages	TypeScript, JavaScript, Python, Rust, Java, C/C++, SQL
Frameworks	React, Next.js, Node.js, Tailwind CSS, Three.js, Flutter
AI/ML	LLM integration, RAG pipelines, MCP, Vercel AI SDK, structured output, LangChain
Infrastructure	Git, Docker, PostgreSQL, gRPC / Protocol Buffers, Vercel, Clerk, Sanity, PostHog, Firebase
Testing & Graphics	Playwright, Vitest, WebGL / GLSL, GSAP, Web Audio API, Web Workers